



Ammune™ User Guide

Version 2

Date: July, 2023

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1 GENERAL

Ammune™ API security solution suite provides **active** and **autonomous protection** from API threats.

It consists of API-Discovery, **API-WAF**, **API-BL**, **API-Bot**, and **API-DDoS** protection modules. It is based on Ammune.ai's proprietary AI algorithm designed to counter cyber security threats.

This document is made to guide users and administrators in Ammune™ version 2.x functionality and usage. Its index follows the menu entries of the Ammune™ console dashboard.

(See 'Main Dashboard')

Website: www.ammune.ai

LinkedIn: <https://www.linkedin.com/company/ammune.ai/>



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2 API Security

2.1 Main Dashboard

The **Main Dashboard** presents a snapshot of the Ammune™ protection activity status.

- **Traffic flow graph:** This tab displays the traffic rate (normal and malicious traffic) during the last 10 minutes in five-second intervals. The number shown is the most recent number of requests in a five-second window
- **Alerts Summary:** This displays the number of alerts per module, identified during the last twenty-four hours

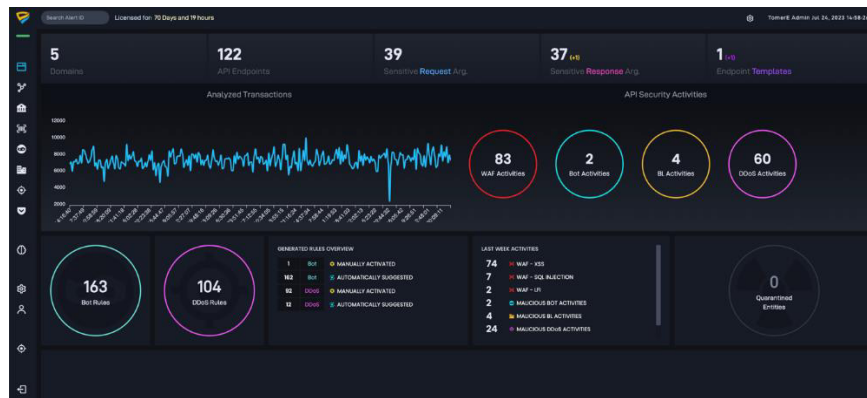


Figure 1: The main dashboard and (left) menu

Clicking the gears icon opens the **Config Change Log** that details changes in Ammune™ configuration.

The screenshot shows the Ammune Config Change Log. It displays a table of configuration changes with columns for Command ID, User, Operation, Details, and Time.

Command ID	User	Operation	Details	Time
10024	joelhu	Edit	Changed System Configuration	May 4, 2022 11:49:10
10025	maelin	Add	Created new protected domain (farcu.com)	Nov 21, 2022 16:24:16
10024	maelin	Edit	Protected domain demotheating.com has been edited	Nov 17, 2022 15:35:22
10024	maelin	Add	Created new protected domain testtheat.com	Nov 16, 2022 17:00:21
10023	william	Add	Added Machine License	Oct 26, 2022 13:58:55
10022	william	Edit	Protected domain theating.com has been edited	Aug 26, 2022 9:54:07
10021	william	Edit	Changed System Configuration	Jul 18, 2022 15:54:35
10020	maelin	Add	Added Machine License	Jul 3, 2022 15:44:39
10019	maelin	Edit	Protected domain demotheating.com has been edited	Jun 15, 2022 14:54:13
10018	william	Edit	Changed System Configuration	Jun 13, 2022 14:54:04
10017	william	Edit	Changed System Configuration	Jun 9, 2022 13:57:57
10016	maelin	Edit	Protected domain theating.com has been edited	Jun 7, 2022 13:07:01
10015	maelin	Edit	Protected domain theating.com has been edited	May 16, 2022 11:16:05
10014	maelin	Edit	Protected domain theating.com has been edited	May 17, 2022 16:54:24
10013	maelin	Add	Created new protected domain theating.com	May 12, 2022 14:12:59
10012	maelin	Add	Added Machine License	May 3, 2022 12:41:32
10011	maelin	Add	Added Machine License	Mar 31, 2022 16:54:47
10010	maelin	Add	Added Machine License	Mar 4, 2022 16:03:19

Figure 2: Configuration Change Log



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2.2 API Discovery

Ammune™ continuously learns and displays the inventory of consistently used **APIs and API endpoints**. API endpoints are grouped by **name** and **method**.

Search Alert ID: [] Licensed for: 75 Days and 2 hours TomerE Admin Jul 20, 2023 8:08:06

API Discovery

All APIs

Filter by API Name

API Name	Total Endpoints	Sensitive Arguments (Requests)	Sensitive Arguments (Responses)	Templates	Covergence Status	First Update	Last Update	New Endpoints	Changed Endpoints	New Sensitive Arguments (Requests)	New Sensitive Arguments (Responses)
demo17testing.com	22	18	29	1	✓	Apr 3, 2022	44d 19h	0	1	0	1
UarcuLcom	14	0	3	0	✓	Nov 21, 2022	240d 18h	0	0	0	0
U7testing7.com	29	8	4	0	✓	Mar 31, 2022	64d 10h	0	29	0	0
U7testing8.com	29	6	3	0	✓	Mar 31, 2022	5s	0	29	0	0
U7testing9.com	29	7	4	0	✓	Mar 31, 2022	10s	0	29	0	0

Figure 3: APIs Inventory

Clicking on an **API**, its **endpoints** are presented with additional data. The date of first and last request, number of requests, response length range, response time range, and detected threat types.

Search Alert ID: [] Licensed for: 68 Days and 20 hours doron Admin Jul 26, 2023 14:03:16

API Discovery

API Endpoints

All APIs > demo17testing.com

Filter by API Endpoint Name

API Endpoint Name	Arguments	Sensitive Arguments (Requests)	Sensitive Arguments (Responses)	Template	Covergence Status	First Update	Last Update	New Sensitive Arguments (Requests)	New Sensitive Arguments (Responses)
PUT /inventory/users	36	5	6		✓	Apr 3, 2022	4d 20h	0	1
POST /inventory/shop/products	34	3	1		✓	Apr 3, 2022	4d 20h	0	0
GET /inventory/shop/products	34	0	3		✓	Apr 3, 2022	190d 12h	0	0
PUT /inventory/shops	30	4	4		✓	Apr 3, 2022	62d 22h	0	0
POST /auth/signin/shop	28	1	1	Login Flow	✓	Apr 3, 2022	35d 0h	0	0
POST /payment/makepay	28	1	0		✓	Apr 3, 2022	47d 19h	0	0
GET /inventory/users	25	0	5		✓	Apr 3, 2022	47d 18h	0	0
PUT /inventory/user/cart	24	1	0		✓	Apr 3, 2022	47d 19h	0	0

Figure 4: API Endpoints Inventory (Per API)

Clicking on an **API endpoint**, its **arguments** are presented, for request and response. The arguments are classified into readable categories (numeric, name, limited text, email, etc.), along with specific length statistics. **Sensitive arguments** are also marked according to the given PII and JWT templates.



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Argument Type	Argument Name	Sensitive	Tags	Description
RESPONSE BODY	user.hash_pass	SENSITIVE	PI token	Characters seen are numbers, letters, punctuation and symbols +4 More
RESPONSE BODY	user.identity_card	SENSITIVE	PI unique identification number	Characters seen are numbers +5 More
RESPONSE BODY	user.address	SENSITIVE	PI address	Characters seen are numbers, letters, punctuation and symbols +4 More
RESPONSE BODY	user.first_name	SENSITIVE	PI name	Characters seen are letters +4 More
RESPONSE BODY	user.last_name	SENSITIVE	PI name	Characters seen are letters +4 More
RESPONSE BODY	user.email	SENSITIVE	PI email	Characters seen are numbers, letters, punctuation and symbols +4 More
REQUEST BODY	first_name	SENSITIVE	PI name	Character's seen are letters +3 More
REQUEST BODY	address	SENSITIVE	PI address	Characters seen are numbers, letters, punctuation and symbols +3 More

Figure 5: API Endpoint argument Inventory

2.3 Sensitive Arguments

Sensitive arguments gathered from all API endpoints are listed here with additional information that is presented at 'API Discovery'. The user can manually add or remove the **sensitive** tag from each argument.

Sensitive arguments are used as part of the Bot 'Generated rules' to determine the auto setting of the policy per endpoint (see 2.5.4).

Domain	API Endpoint	Argument Type	Argument Name	Sensitive	Tags	Description
demo7testing.com	POST /inventory/shop	RESPONSE BODY	shop_email	SENSITIVE	PI email	Characters seen are numbers, letters, punctuation and symbols +4 More
demo7testing.com	POST /inventory/shop	RESPONSE BODY	shop_name	SENSITIVE	PI name	Characters seen are letters +3 More
demo7testing.com	POST /inventory/shop	RESPONSE BODY	address	SENSITIVE	PI address	Characters seen are numbers, letters, punctuation and symbols +3 More
demo7testing.com	POST /inventory	REQUEST URL	access_token_exp	SENSITIVE	PI token	Characters seen are numbers +4 More
demo7testing.com	POST /inventory	REQUEST URL	access_token_exp	SENSITIVE	PI token	Characters seen are numbers +4 More
demo7testing.com	DELETE /inventory/user/cart	REQUEST URL	object_id	SENSITIVE	PI token	Characters seen are numbers +4 More
demo7testing.com	POST /inventory/user/cart	REQUEST BODY	cart_number	SENSITIVE	PI unique identification number	Characters seen are numbers +4 More
demo7testing.com	POST /inventory/user/cart	REQUEST URL	object_id	SENSITIVE	PI token	Characters seen are numbers +4 More
demo7testing.com	POST /inventory/users	RESPONSE BODY	last_name	SENSITIVE	PI name	Characters seen are letters +3 More
demo7testing.com	POST /inventory/users	RESPONSE BODY	address	SENSITIVE	PI address	Characters seen are numbers, letters, punctuation and symbols +3 More
demo7testing.com	POST /inventory/users	RESPONSE BODY	first_name	SENSITIVE	PI name	Characters seen are letters +3 More
demo7testing.com	POST /inventory/users	RESPONSE BODY	email	SENSITIVE	PI email	Characters seen are numbers, letters, punctuation and symbols +4 More

Figure 6: Sensitive Arguments Inventory

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2.4 API-WAF

2.4.1 Real Time Dashboard

The real time dashboard presents live traffic, last generated alerts (blocked requests), and their geo-origins on a map.

- **Traffic flow graph:** The traffic rate (normal and malicious traffic) during the last 10 minutes in five seconds intervals
- **Latest Alerts:** A list of the most recent alerts
- **Malicious IPs summary:** A map that presents the malicious IP origins

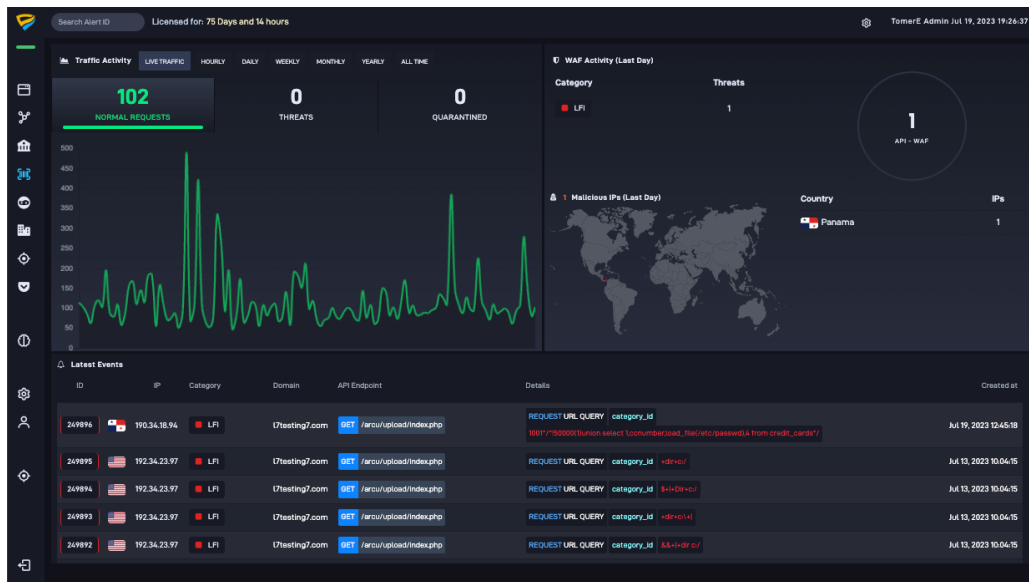


Figure 7: API-WAF Real Time Dashboard

2.4.2 Alerts

2.4.2.1 Alerts Groups

The last 10-minute alerts are presented in level 1 groups (see below).

The user can select criteria for Alerts retrieval. The selection options are:

- **Time:** Last 10 Minutes, last Hour, Day, Week, Month, or All-time
- **Domain:** ALL, specific domain name
- **Attack Category:** ALL, SQL injection, Log4J injection, etc
- **Method:** HTTP method used – ALL or specific methods (GET, POST, etc)
- **ARG Type:** ALL, or specific request part (Header...)

Maximum of 50,000 alerts can be retrieved.



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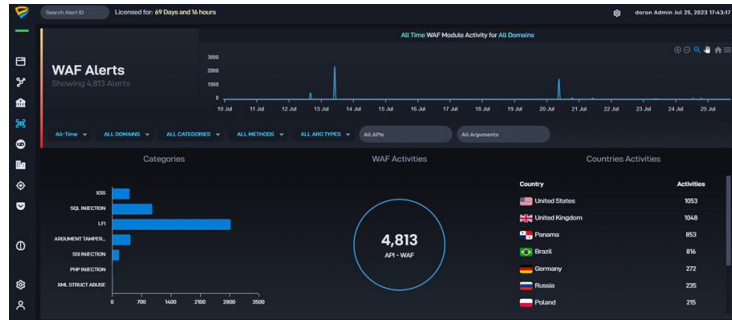


Figure 8: API-WAF Alerts - Part1 (Up)

Alerts are presented in a two-level group model:

- **Level 1** - alert group characteristics {Attack type + API Endpoint + argument}
- **Level 2** - alerts sub-group characteristics {Level 1 + signature}
- * Signature - attack vector features

Clicking on a (layer 1) group will unfold to present the (layer 2) sub-groups



Figure 9: API-WAF Alerts - Part 2 (Down – Layer 1 groups)

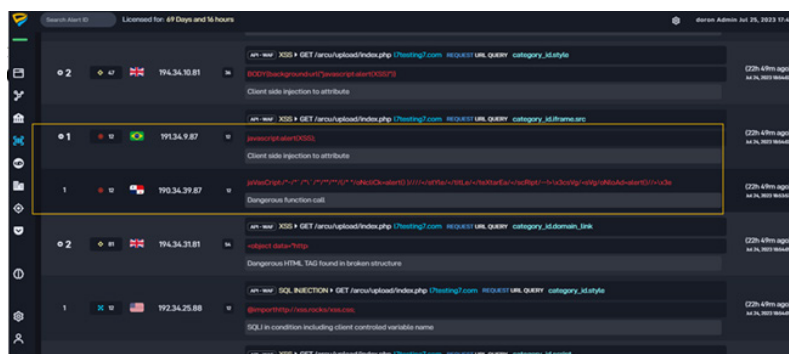


Figure 10: API-WAF Alerts - Part 2 (Layer 2 sub-groups)



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2.4.2.2 Alerts Reports

The WAF Alerts page also contains an option to generate **reports**. Clicking on the '**Download Report**' button generates a report that reflects the current user alert selection. The reports have two parts: **alert summary** and **details**.

Alerts by Category Showing 3 categories		Jul 24, 2023 13:10:08 Last Day
Category	Description	Number of Alerts
XSS	1) Dangerous HTML TAG found in broken structure. Sign of XSS attempt 2) Client side injection to attribute. When argument embedded in HTML attribute it appends malicious payload, triggering XSS 3) Dangerous function call found. It can be either client side (XSS) or server side function call (php/Asp injection)	74
SQL INJECTION	1) SQLi injection to string. For example look at select a,b,c from table1 where a='\$val'. If val="foo" or 1=1" the query changes to select a,b,c from table1 where x='foo' or 1=1 2) SQLi in condition including client controlled variable name. Example: select a,b,c from table1 where a.price=b.\$val. In this case the following injection may work "foo; select * from table1". There is not need in quotation mark to start injection	7
LFI	1) Directory traversal or local file inclusion found - restricted files	2

Figure 11: API-WAF Alerts report - summary

Alerts Showing 20 results					Jul 24, 2023 13:10:08 Last Day
Vectors	Alerts	IPs	Details	Time	
1	1	 190.34.9.97	XSS [GET /arcu/upload/index.php] I7testing7.com [req url query] [category_id.foo]	Jul 24, 2023 10:48:02	
1	1	 190.34.9.97	XSS [GET /arcu/upload/index.php] I7testing7.com [req url query] [category_id.name],); phpinfo(); exit;/*	Jul 24, 2023 10:48:02	
1	1	 190.34.9.97	SQL INJECTION [GET /arcu/upload/index.php] I7testing7.com [req url query] [category_id.foo] or 1=1 or =	Jul 24, 2023 10:48:01	
1	2	 190.34.12.94  191.34.46.84	LFI [GET /arcu/upload/index.php] I7testing7.com [req url query] [category_id.domain_link] c:\program files\adobe\acrobat 7.0\acrobat\acrobat.dll	Jul 24, 2023 10:48:00	
2	8	 190.34.12.94  190.34.13.81  194.34.37.87  194.34.9.93	XSS [GET /arcu/upload/index.php] I7testing7.com [req url query] [category_id.style] BODY{-moz-binding:url("http://xss.rocks/xssmoz.xml#xss")}	Jul 24, 2023 10:47:59	

Figure 12: API-WAF Alerts report - details



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2.4.2.3 Alerts Cluster

Clicking on a specific alerts group (level 2), open the **'general attacking group'** with a summary of information about the alerts group and detailed information per alert:

- **Target data:** API endpoint and argument
- **Attack data:** Attack source, attack category, attack vector, and summary

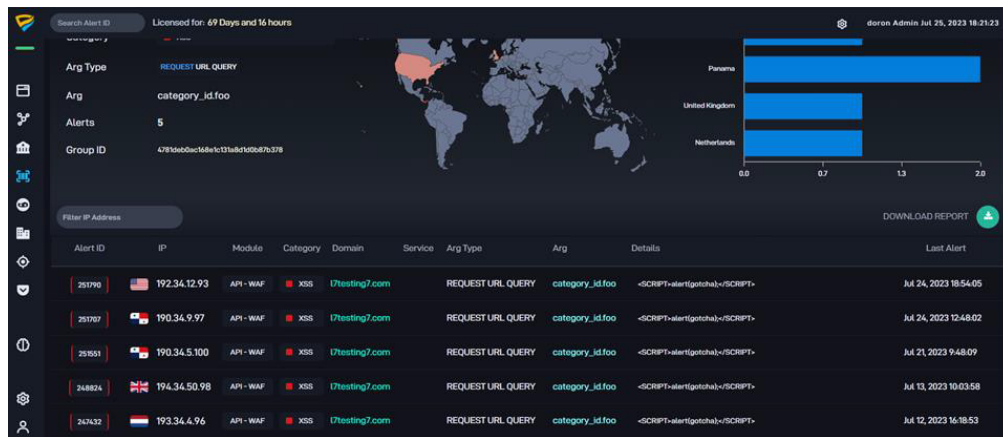


Figure 13: API-WAF Alerts cluster

2.4.2.4 Alert Details

Detailed alert data is presented as well as a list of similar alerts. A **traffic sample** is also presented of the HTTP header and a zoom-in to the specific malicious attack pattern nested inside the request.

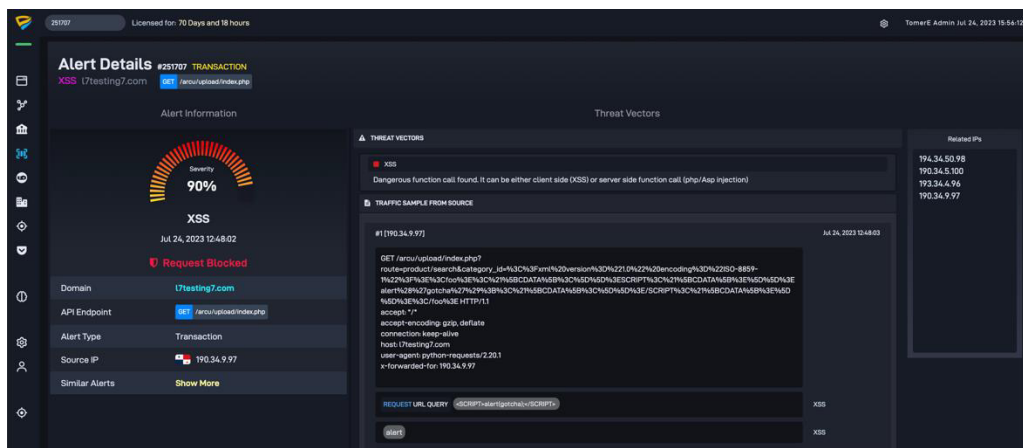


Figure 14: WAF Alert details



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2.4.2.5 Exclusions (from the "Alerts" View)

Exclusion rule from further being blocked can be set manually for a specific alerts group, directly from the 'Alerts View' (2.4.2.1), by clicking on the 'Edit Exclusions' slider.

'Enable Similar Request' (green button) is lit on and used to disable the specific alerts group with a single click.

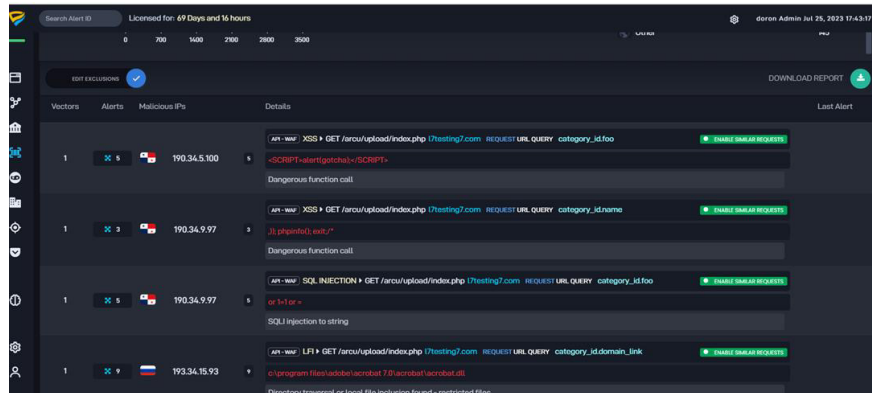


Figure 15: Direct alerts exclusion at the API-WAF Alerts

2.4.3 Exclusion

Exclusion rules may be set directly from the 'Exclusion' menu entry.

In this case, the exclusion can be applied according to more robust selection criteria.

- **Pick Exclusion:** Creating a new exclusion rule or editing an existing one
- **Domain:** The domain that will be associated with the exclusion rule
- **Method:** The request method
- **Category:** The threat type
- **Argument Type:** header, body...
- **Argument:** Ammune™ will detect according to argument names. Custom argument names are also supported

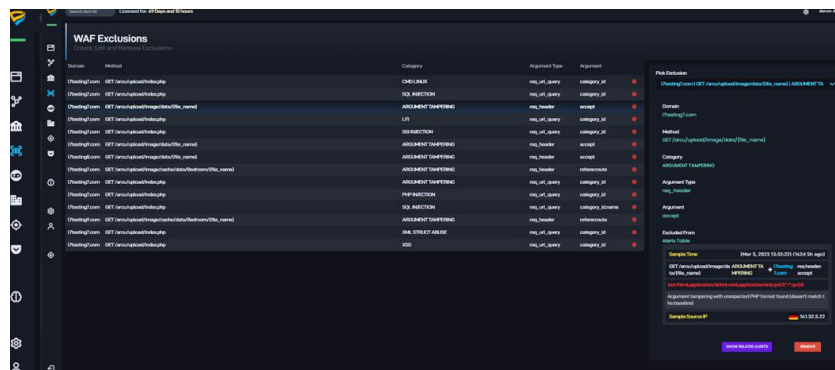


Figure 16: API-WAF Exclusions



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2.4.4 Malicious IPs

Most recent malicious IPs that were quarantined are presented. It is possible to refine the list according to the IP address and age (10 minutes, hour, day, week, month, no limit), or only by the IP address.

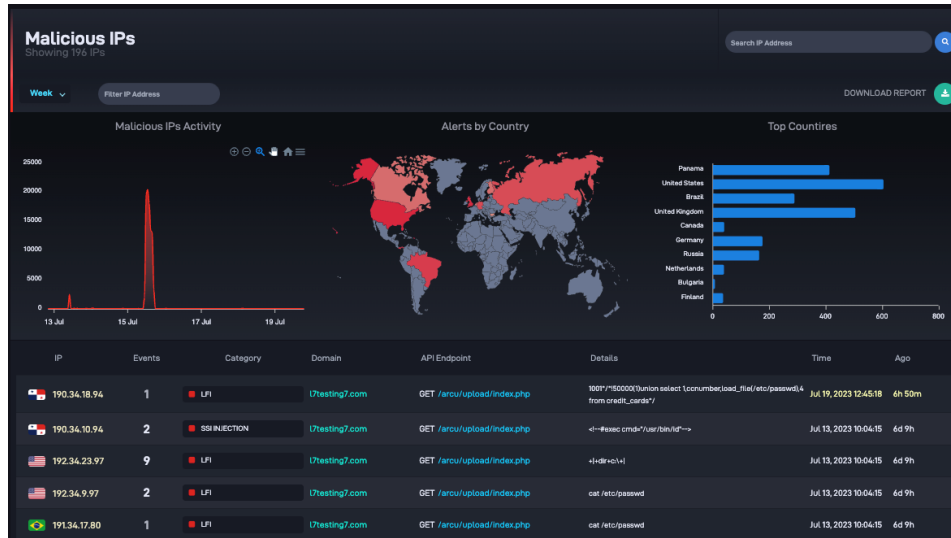


Figure 17: Malicious IPs

2.4.5 API-WAF Policy

Available WAF protection policies are listed here. Clicking an existing policy row, the detailed policy form is opened (on the right).

Create a new policy from the topmost field (Pick Policy). The protection policy controls the protection activity as follows:

- **Policy name:** Chosen by the user
- **Block Malicious Requests:** Ammune™ will block or monitor malicious requests by this policy
- **Suggested Quarantine Time:** This parameter determines the blocking duration in seconds of malicious IP sources
- **Use AI Quarantine Time:** This allows Ammune™ to make an autonomous decision on quarantine time
- **Disable Header Errors:** This option disables the conformance checks for the configured types of HTTP header errors
- **Disable Body Errors:** This option disables conformance checks (or parsing of error alerts) for the configured types of HTTP body errors
- **Inspect Body:** This option controls whether the injection inspection is done at the request argument body
- **Inspect Arg Names:** This option controls whether WAF injection inspection is done on argument names. It is relevant to URL, Body, Cookie, XML attribute, and header argument names



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- **Inspect Cookies:** This option controls whether injection inspection is done on cookie request arguments
- **Inspect Headers:** This option controls whether injection inspection is done on HTTP HEADER request arguments
- **Inspect URL Arg:** This option controls whether injection inspection is done on URL request arguments
- **Inspect URL Path as Arg:** This option controls whether injection inspection is done on the URL path
- **Block Any File Upload:** When set to **ON**, this option disables uploading files. By default, this option is set to **OFF**

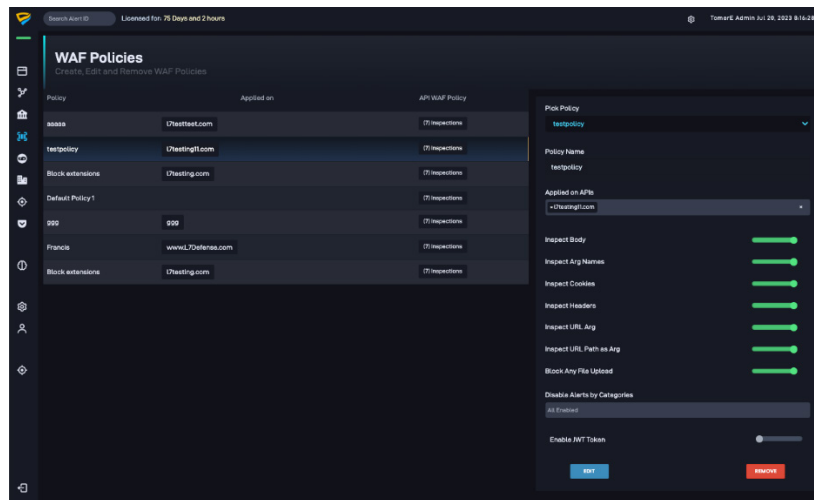


Figure 18: API WAF POLICY SETTINGS

Note: Special caution must be taken when disabling errors, especially body and URL parsing errors. The result is skipping inspection of fields that cannot be retrieved. It is strongly recommended to resolve conformance issues in the client application instead.



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2.5 API-BOT

2.5.1 Attacking Bots & Alerts

Bot alerts are generated according to '**Bot Rules**' (2.5.3) that are specific to each API endpoint. Filters – **time frame**, **entity type** (source IP, token, or **user**), and **domain**, can be applied to select the subgroup of alerts. Bot activity is auto-tagged as 'Excessive Usage' (per multiple targets), 'Enumeration' (per multiple targets), and '**Brute Force**' (per a single target). In cases an alert doesn't fall into these categories, it is not tagged.

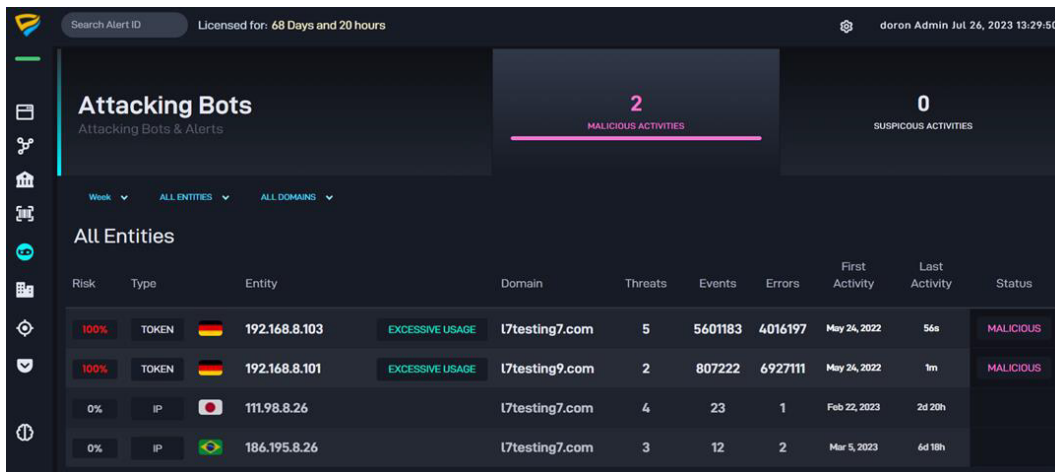


Figure 19: Bot alerts

2.5.2 Bot Alert (Full Entity Identifier)

Clicking on a **bot alert** opens the **Full Entity Identifier**, which contains information related to the specific bot activity – first and last appearance, list of targeted API Endpoints, **responses** statistics, and more.

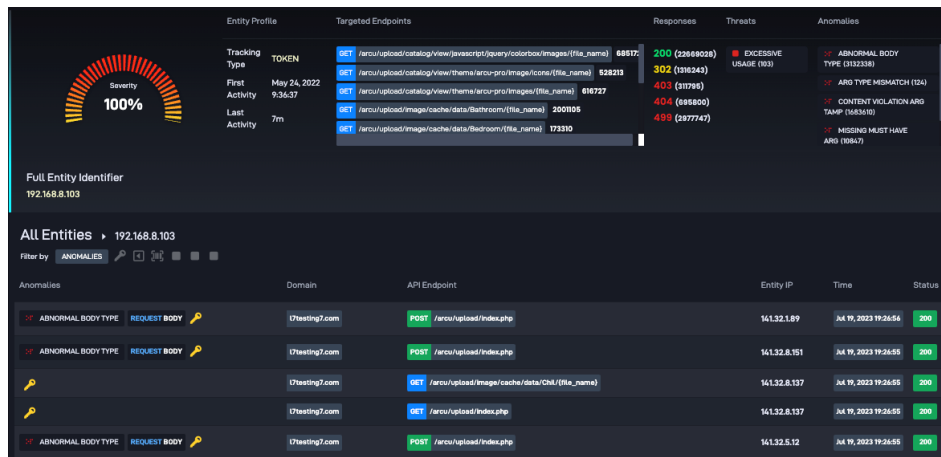


Figure 20: Bot alerts

2.5.3 Generated Bot Rules

Bot rules are generated automatically for each and every API endpoint that Ammune™ discovers. Bot rules are made of multiple time resolution values for normal traffic rate, as well as other calculated features. API endpoint with sensitive arguments is automatically set as **active** (if enforcement is on, the rule will be used to start blocking). Users can control the state of the rules via the 'delete all', 'apply all', and 'reset rules' buttons.

By clicking on the specific API endpoint users can adjust the traffic rates that trigger alerts to other values. However, it is not recommended, unless for a very good reason.

Figure 21: Bot rules

2.5.4 Bot Settings

Bot setting enables the user to adjust the period a hostile source will be quarantined. The first choice when the slider is on (Default), is that the AI will choose the period.

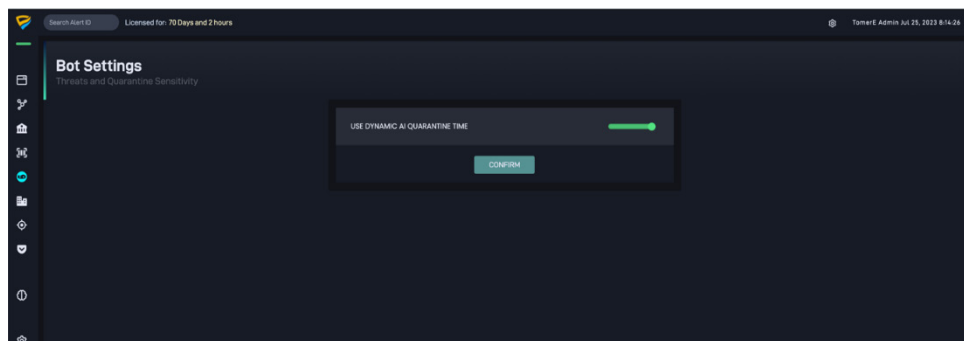


Figure 22: Bot Settings (slider is on)



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As the slider turns **off**, the user can decide upon the time periods for the user, token, and source IP.

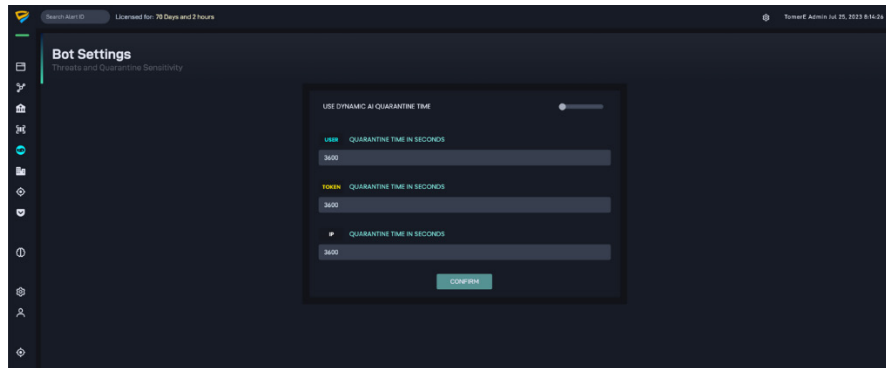


Figure 23: Bot Setting (Slider is off)



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2.6 API-BL

2.6.1 BL Attackers & Alerts

Business logic alerts are generated according to encoded **BL Rules**. Filters – time frame, entity type (source IP, token, or user,) and domain, can be applied to select subgroups of alerts. The BL attack is auto-tagged according to the bold part of the attack (Endpoint tampering, content violation, and more).

Risk	Type	Entity	Domain	Threats	Events	Errors	First Activity	Last Activity	Status
70%	IP	192.34.16.87	l7testing7.com	2	21	3	Jul 20, 2023	6d 5h	MALICIOUS
70%	IP	191.34.34.95	l7testing7.com	3	129	8	Jul 20, 2023	6d 5h	MALICIOUS
80%	IP	190.34.23.80	l7testing7.com	3	135	14	Jul 20, 2023	6d 5h	MALICIOUS
88%	IP	191.34.18.99	l7testing7.com	3	142	20	Jul 20, 2023	6d 5h	MALICIOUS

Figure 24: BL Alerts

2.6.2 BL Alert

Clicking on a **BL alert** opens the **Full Entity Identifier**, which contains information related to the specific user activity – first and last appearance, list of targeted API Endpoints, and related anomalies.

Category	Domain	API Endpoint	Entity IP	Time	Status
CONTENT VIOLATION SSR	l7testing7.com	/frcu/upload/index.php	192.34.16.87	Jul 20, 2023 8:41:42	AB3
CONTENT VIOLATION SSR	l7testing7.com	/frcu/upload/index.php	192.34.16.87	Jul 20, 2023 8:41:42	AB3
CONTENT VIOLATION SSR	l7testing7.com	/frcu/upload/index.php	192.34.16.87	Jul 20, 2023 8:41:42	AB3
CONTENT VIOLATION SSR	l7testing7.com	/frcu/upload/index.php	192.34.16.87	Jul 20, 2023 8:41:42	AB3

Figure 25: BL Alerts



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2.6.3 BL Settings

BL settings include the possibility to set **Threat Sensitivity**, **Quarantine Sensitivity**, and use **Dynamic AI Quarantine Time** (see 2.5.3)

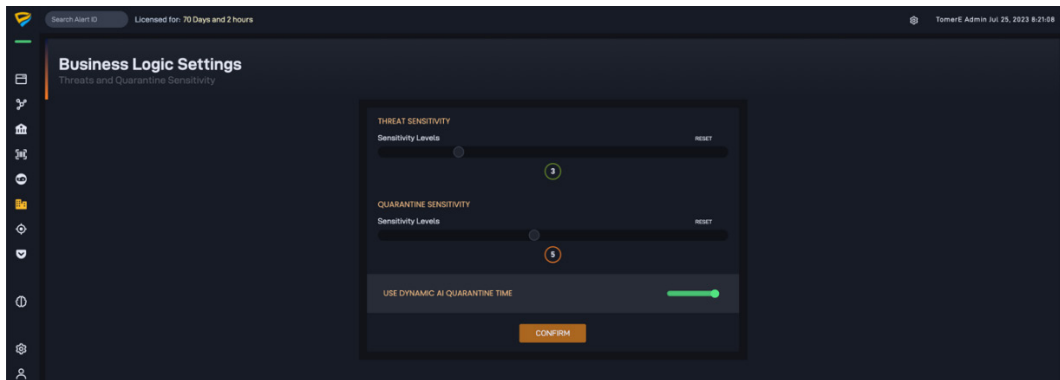


Figure 26: BL Setting (Slider on)

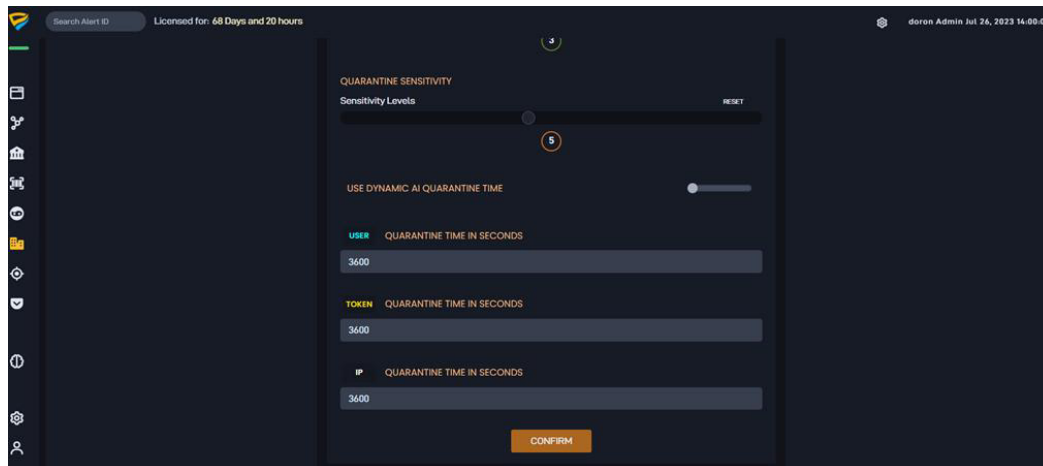


Figure 26: BL Setting (Slider off)



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2.7 API-DDoS

2.7.1 DDoS Attackers & Alerts

DDoS alerts are generated according to "DDoS Rules" (2.7.3) that are specific to each API endpoint. Filters – time frame, entity type (source IP, token, or user,) and domain can be applied to select subgroups of alerts.

Risk	Type	Entity	Domain	Anomalies	Events	Errors	First Activity	Last Activity	Status
100%	IP	50.418.26	l7testing7.com	2	44	6	Jul 26, 2023	3h 35m	MALICIOUS
81%	IP	36.145.8.26	l7testing7.com	2	37	3	Jul 26, 2023	6h 23m	MALICIOUS
79%	IP	200.51.8.26	l7testing7.com	2	18	6	Jul 26, 2023	9h 49m	MALICIOUS
88%	IP	92.145.8.26	l7testing7.com	2	39	6	Jul 25, 2023	22h 4m	MALICIOUS
81%	IP	122.253.8.26	l7testing7.com	2	37	3	Jul 25, 2023	1d 6h	MALICIOUS
76%	IP	61.105.8.26	l7testing7.com	3	21	6	Jul 25, 2023	1d 6h	MALICIOUS
84%	IP	156.249.8.26	l7testing7.com	2	34	6	Jul 25, 2023	1d 8h	MALICIOUS

Figure 28: DDoS attackers & Alerts

2.7.2 DDoS Alert

'DDoS alert' is similar to 'Bot alert' (See 2.5.2)

Anomalies	Domain	API Endpoint	Entity IP	Time	Status
ABNORMAL BODY TYPE	l7testing7.com	POST /arou/upload/index.php	50.418.26	Jul 26, 2023 10:19:43	200
ABNORMAL BODY TYPE	l7testing7.com	POST /arou/upload/index.php	50.418.26	Jul 26, 2023 10:19:42	200
ABNORMAL BODY TYPE	l7testing7.com	POST /arou/upload/index.php	50.418.26	Jul 26, 2023 10:19:42	200

Figure 29: DDoS Alert



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2.7.3 Generated DDoS Rules

'DDoS rules' is similar to 'Bot rules' (See 2.5.3)

Entity Type	Domain	API Endpoint	Traffic Rates	Status	Created At	Last Update
IP	Darcu.com	GET /arcu/upload/catalog/view/theme/default/image/[file_name]	3 3 12 12	ACTIVE	Nov 21, 2022	19d 3h
IP	Darcu.com	GET /arcu/upload/image/cache/data/texture/[file_name]	3 6 24 48	ACTIVE	Nov 21, 2022	55d 2h
IP	Darcu.com	POST /arcu/upload/index.php	3 12 48 96	SUGGESTED	Nov 21, 2022	20d 13h
IP	Darcu.com	GET /arcu/upload/index.php	3 12 96 192	SUGGESTED	Nov 21, 2022	20d 13h
IP	Darcu.com	GET /arcu/upload/image/cache/data/child/[file_name]	3 12 24 96	SUGGESTED	Nov 21, 2022	20d 13h
IP	Darcu.com	GET /arcu/upload/image/cache/data/bedroom/[file_name]	3 24 96 96	SUGGESTED	Nov 21, 2022	20d 13h
IP	Darcu.com	GET /arcu/upload/image/cache/data/bathroom/[file_name]	3 6 96 96	SUGGESTED	Nov 21, 2022	20d 13h
IP	Darcu.com	GET /arcu/upload/image/cache/data/forest%20space/[file_name]	3 3 12 24	SUGGESTED	Nov 21, 2022	20d 13h

Figure 30: Generated DDoS rules

2.7.4 DDoS Settings

DDoS Settings decides on the list of API endpoints that are sensitive to applicative DDoS, according to the following rules:

- **The Minimum Response Time for Sensitive API Endpoint:** API Endpoints with higher response time will be considered as sensitive to DDoS
- **Minimum Response Size for Sensitive API Endpoint:** API Endpoints with higher response size will be considered as sensitive to DDoS
- **Minimum Number of Objects for Sensitive API Endpoint:** API Endpoints with a higher number of objects will be considered as sensitive to DDoS

Figure 31: DDoS Setting

The user can also define the quarantine period for Bots (2.5.4)



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2.8 Manage Entities

'Manage Entities' controls users and system-defined white and black lists of entities, i.e., IP, User, Token, including quarantined entities.

The options are:

- **User-defined whitelist and blacklist lists of entities:** Clicking on an entity corresponding line, an editing form is opened
- **User-defined entity list that do not quarantine:** The same method to register entities is applied, as described above
- **Algorithm-defined temporarily quarantined entities list:** Decided automatically by one or more of Ammune™ modules. Clicking on an action option, move the entity to one of the above lists (user action)

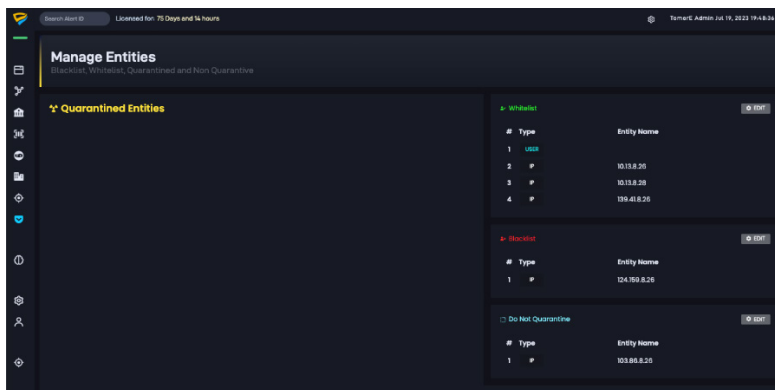


Figure 32: Manage Entities



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2.9 Settings

2.9.1 Proxy Setting

A new traffic forwarding rule is created by selecting '**Create New**' in the '**Pick Protected Domain**' option, on the form on the right side (default option). Fields in this form are mandatory. Once complete, click **Create**. Each domain/forwarding rule entry specifies the following:

- **Domain:** The protected domain
- **Front IP:** The IP that Ammune™ receives the inbound traffic
- **Domain Port:** The port that Ammune™ receives the inbound traffic
- **Domain Protocol:** The protocol that Ammune™ accepts (HTTP or HTTPS) o HTTPS : See Setting - 'TLS certificates'
- **Protected Instance and Protected Instance Port:** The IP and port of the protected backend server, which Ammune™ forward requests to, and receives replies from
- **Protected Instance Protocol:** Forwarding request traffic type (HTTP or HTTPS)
- **Enable Health Check:** This field is used to apply health checks to Ammune™ itself for the specific proxy line (domain name) If this flag is on an additional text box will appear with a URL from which Ammune™ will reply with health verification

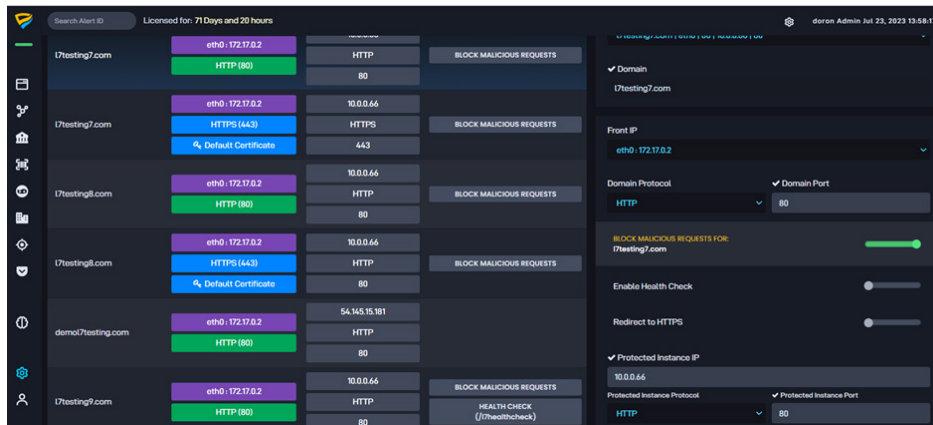


Figure 34: Proxy Settings



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Note:

1. The format of the forwarding rule is shown below:

`<domain name, forward_ip:forward_port, protocol> -> <backend_ip:backend_port, protocol>.`

2. The domain name entered in the reverse proxy configuration may not include any wildcards. Internally, reverse proxy relates both `*.<domain_name>` and `<domain_name>` to the same rule. The longest match rule is selected when several forwarding rules match an actual host domain name (HTTP host header) on the given port.

3. When a reverse proxy rule with `<domain_name>` is created, a default domain profile is generated for that `<domain_name>`. It is possible to edit this profile in the domain profiles screen. However, subdomain profiles are not created automatically and must be added manually if non-default profile settings are required.



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2.9.2 Global Settings

Here, the global setting of the solution is defined. The options are:

- **System mode:** Supporting two deployment models – single server and cluster of real-time agents (engines) vs. central management & analytics unit.

The deployment options per server are:

- All in one (Single server)
- Management only (Management & Analytics)
- Engine only (Real-time unit)

The last two options are used in a cluster deployment

- **Engine mode:** the deployment model of the solution – Inline, Monitor
- **Proxy mode:**
 - DNS directly - Traffic is routed directly to the Ammune™ server
 - Behind the load balancer (Default) - Traffic is routed to a load balancer (LB) that embeds real customer IP in the header
- **Default IP Header (behind LB):** The header of the LB that encodes the real source IP
- **IP Headers from trusted proxies:**
 - **Header:** Define X-forward from
 - **Accept from IP Range:** Relevant IP range (Upstream LB)

Google Cloud (GCP) exception: Google LB internal IP range and specific LB IP address must be specified as Proxies, as the X-forward can't be used. In GCP, the load balancer adds two IP addresses to the X-Forwarded-For header (X-Forwarded-For: <true_client_ip> and an X Forwarded-For: <lb_ip>), therefore the **Behind-the-LB** mode doesn't work.

- **Time zone:** Set the time zone

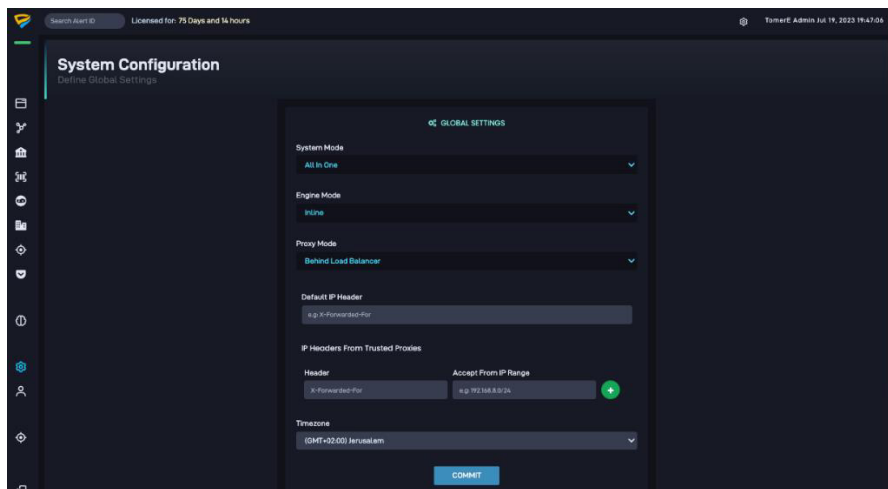


Figure 35: Global Settings



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2.9.3 TLS Certificates

TLS certificates are uploaded here for the usage of the 'proxy setting' configuration

- **Pick certificate:** Existing for an update or adding a new one
- **Certificate name:** User defined
- **Private key password (optional):** if no key password, leave empty
- **Private key:** .pem or .key format are supported
- **Public certificate:** Certificate to be uploaded. crt/key certificate format is supported

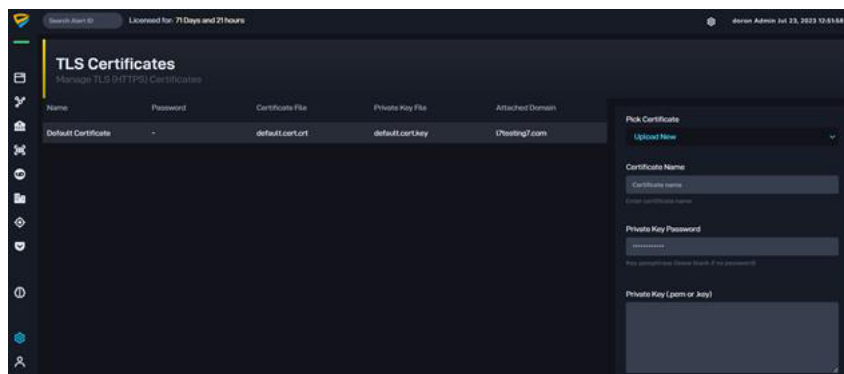


Figure 36: TLS Certificates



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2.10 Admin

2.10.1 UI-Users

Users are added or removed using the **UI Users**.

A user may be defined as an admin or observer.

- **Pick UI User:** Existing for an update or new
- **Username:** Chosen by the user
- **Email**
- **User Level:** Observer or admin
- **Password / Confirm Password:** User password

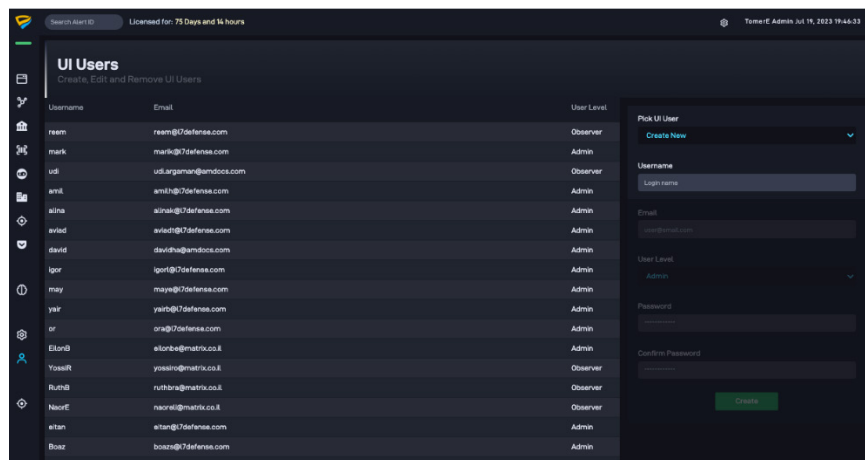


Figure 37: UI Users View



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2.10.2 Machine License

A license determines the combination of modules enabled on a specific solution and the license period. The license can be added or updated by 'copy-paste' of the license key.

Machine License
Add License or Credits

LICENSED

17 Ammun version: v1.09.1.08801

Active Modules: API - WAF API - BOT API - BL API - DDoS

Machine ID: da33054b907787c325334c4836f0e0

Expires In: 75 Days and 14 hours

Current License
DQwR5A2QZ7f7L/C98B4Rw4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r
H6pG0g0g8Tm0P0u0a0u00400400B0Mv02U0u0u0U0S/3Mv5HE0q0u0yA0h0d0g0u0v+0ach0e0g0h0M3u0k3
f0Ag022U0u0B0u0d0e008f0m0m0y0C0C40480q0g0e0C0A0E0M0Z0M0S/+00W0N0E00w0C00X0C0u0d0n0g0Q==

ISSUED LICENSES

License	Modules	Expires In
DQwR5A2QZ7f7L/C98B4Rw4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - WAF	75 Days and 14 hours
8J0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - BOT	
9d0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - BL	
0u0u0U0S/3Mv5HE0q0u0yA0h0d0g0u0v+0ach0e0g0h0M3u0k3	API - DDoS	
8J0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - WAF	75 Days and 14 hours
9d0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - BOT	
0u0u0U0S/3Mv5HE0q0u0yA0h0d0g0u0v+0ach0e0g0h0M3u0k3	API - BL	
8J0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - DDoS	
8J0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - WAF	75 Days and 14 hours
9d0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - BOT	
0u0u0U0S/3Mv5HE0q0u0yA0h0d0g0u0v+0ach0e0g0h0M3u0k3	API - BL	
8J0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - DDoS	
8J0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - WAF	75 Days and 14 hours
9d0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - BOT	
0u0u0U0S/3Mv5HE0q0u0yA0h0d0g0u0v+0ach0e0g0h0M3u0k3	API - BL	
8J0P4nrcYAB7N4TDC3/NTA9O5P3ta4nY846Shu7qHngBCK8YV0abhdApKic3r	API - DDoS	

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Figure 38: Machine license



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